AI-Based Virtual Try-On & Size Recommendation System

1. Component Overview

The AI-Based Virtual Try-On and Size Recommendation system operates as a comprehensive digital tailor and virtual fitting room. It utilizes a cascade of predictive Machine Learning models to autonomously estimate a customer's precise physical body measurements based solely on basic inputs (height, weight, and gender), eliminating the need for manual tape measurements. Furthermore, it employs a classification model to translate standard sizes into brand-specific sizing, accounting for manufacturer discrepancies. Finally, it integrates Generative Artificial Intelligence to digitally composite garments onto the customer's body, providing a photorealistic preview of the fit and appearance.

2. Problem Addressed

A persistent challenge in the modern retail and e-commerce industry is clothing size inconsistency across different brands, known colloquially as "vanity sizing." A customer who wears a "Medium" in one brand may require a "Large" or "Small" in another, leading to incorrect purchases, exceptionally high return rates, and customer dissatisfaction. Additionally, customers often lack the confidence to purchase garments without physically trying them on, a process which is time-consuming in physical stores and impossible in digital contexts.

3. Main Objective

The primary objective of this component is to develop an integrated AI-driven ecosystem that accurately predicts a customer's physical body measurements, intelligently recommends brand-specific clothing sizes to mitigate cross-brand discrepancies, and provides a photorealistic virtual try-on experience to enhance purchasing confidence.

4. Specific Objectives

- To train and deploy a multi-output regression Machine Learning model capable of predicting detailed body measurements (shoulder width, waist, leg length) using only height, weight, and gender.
- To engineer an automated brand-sizing classifier that maps standard sizes to brand-specific proprietary sizes based on clothing category.
- To integrate a Generative AI pipeline capable of performing image-to-image garment re-contextualization (Virtual Try-On).
- To construct an intuitive mobile user interface that seamlessly captures user metrics and renders high-fidelity generated images.

5. Main System Functions

- Standard Sizing Engine: A rule-based baseline calculator that derives a standard garment size (S, M, L, XL) from height and weight indices.
- Body Measurement Prediction: Autonomous AI estimation of shoulder width, waist circumference, and leg length.
- Brand-Specific Size Mapping: Real-time, on-demand prediction of the exact size label required for a specific brand's product.
- Generative Virtual Try-On: Photorealistic compositing of a selected garment image onto a user-uploaded photograph, respecting natural body pose and fabric drape.

6. System Workflow

The operational workflow follows a sequential, multi-stage pipeline:

1. Profile Creation: The customer inputs height, weight, and gender into the mobile application.

2. Measurement Inference: The backend calculates BMI and feeds the data into the Random Forest Regressor to predict exact body measurements, storing them in the database.

3. Product Browsing: When the customer views a product, the backend executes the Random Forest Classifier, converting the customer's standard size into the product's brand-specific size.

4. Try-On Request: The customer uploads a personal photo. The system transmits this photo alongside the product image to the Generative AI cloud platform.

5. Image Generation: The Generative AI model processes the images, adapting the garment to the customer's physical geometry.

6. Delivery: The rendered image is returned to the mobile app and displayed alongside the AI-recommended brand size.

7. System Architecture

The system architecture integrates traditional web services with cloud-based AI:

• Client Tier: A mobile application for user data entry and image uploading.

• Middle Tier (FastAPI): A Python backend hosting the pre-trained Scikit-Learn models for immediate inference of measurements and sizes. It also acts as a secure reverse-proxy for the Generative AI service.

• Cloud AI Tier: Google Cloud Vertex AI, hosting the heavily parameterized "virtual-try-on-001" model for image synthesis.

• Data Tier: A NoSQL database persisting user profiles, inferred body measurements, and product catalog metadata.

8. Technologies Used

- Frontend / Mobile App: Flutter (Dart), Image Picker libraries
- Backend API: Python, FastAPI REST framework
- Predictive Machine Learning: Scikit-Learn, Pandas, GridSearchCV (Hyperparameter optimization)
- Generative AI: Google Cloud Vertex AI (virtual-try-on-001)
- Database: Firebase Firestore (NoSQL)

9. Dataset and Data Scope

The predictive models rely on two distinct datasets:

• Anthropometric Dataset: Used for the body measurement model. It contains validated real-world measurements (height, weight, gender, shoulder width, waist, leg length) serving as ground truth. Features were normalized, and BMI was engineered as a synthetic feature.

• Brand Sizing Dataset: Used for the brand-sizing model. It maps standard sizes across various brands and categories to their respective proprietary size labels.

Both datasets were split using an 80/20 train-test distribution.

10. Backend Implementation

The backend services are modularized by AI task:

• body_measurement_service.py: Executes the Random Forest Regressor. It handles the dynamic calculation of BMI and scales inputs for the model.

• brand_sizing_service.py: Executes the Random Forest Classifier. It relies on four serialised Label Encoders to securely map categorical strings (e.g., "Zara", "clothing") into numerical representations for the model.

• gemini_service.py: Manages the asynchronous REST calls to Google Vertex AI. It handles multipart/form-data image payloads and implements safety filtering validation.

11. Frontend Implementation

The frontend implementation focuses on frictionless user experience:

• Profile Management: Captures physical metrics with dynamic unit conversion (cm/inches, kg/lbs) to maximize accessibility.

• Product Detail View: Intercepts the product's brand metadata and asynchronously fetches the AI brand-size prediction before rendering the UI.

• Try-On Interface: Manages dual-image state (human photo and garment photo), handles loading states during the multi-second Generative AI inference, and safely renders the returned URL while bypassing browser cache mechanisms.

12. Database Implementation

The NoSQL database implementation is streamlined to a primary collection:

• Profiles Collection: Functions as a comprehensive digital twin for the customer. It stores standard inputs alongside the AI-predicted outputs (predicted_shoulder_width, predicted_waist, predicted_leg_length). Any mutation to height or weight triggers a transactional recalculation of all predicted fields.

13. Implementation Results

The implementation successfully unified predictive and generative artificial intelligence into a cohesive digital fitting room. The regression model proved capable of extrapolating complex body geometries from minimal inputs, while the generative pipeline successfully composited garments with high photorealism, correctly respecting user posture and garment physics. The architecture demonstrated low latency for numerical predictions and acceptable latency for the heavy image-generation tasks.

14. Machine-Learning Results

- Body Measurement Regressor: Utilizing a Random Forest Regressor optimized via GridSearchCV, the model achieved high R^2 scores and low Mean Absolute Error (MAE), indicating strong correlation between basic metrics and complex body geometry.
- Brand Sizing Classifier: The Random Forest Classifier successfully achieved high categorical accuracy, effectively normalizing erratic brand sizing conventions into a predictable mapping.
- Generative AI: The Vertex AI virtual-try-on-001 model demonstrated excellent zero-shot capability, successfully merging unseen human photos with unseen garment photos without requiring fine-tuning.

15. Important Features Identified by the Models

- Regression Features (Body Measurements): The engineered Body Mass Index (BMI) feature proved highly influential in accurately predicting waist circumference,

while height served as the primary node for leg length splits.

- Classification Features (Brand Sizing): The "Brand Name" categorical feature possessed the highest Gini importance, confirming the hypothesis that brand identity is the primary driver of sizing variance.

16. Functional Testing Results

- Measurement Accuracy: Profile mutations successfully triggered backend recalculations, yielding realistic body measurements across a spectrum of diverse height/weight inputs.

- Brand Mapping: The system correctly translated a standard "Medium" to an "EU 38" or "UK 10" depending strictly on the selected product's brand metadata.

- Visual Compositing: The Generative AI effectively handled complex occlusions, such as arms crossing the torso, realistically rendering the garment around the user's limbs.

17. Research Contribution

This research contributes a novel, cascaded AI architecture that bridges the gap between statistical predictive modeling and Generative AI in the retail sector. By establishing that precise anthropometric measurements can be reliably inferred from trivial inputs (height/weight), the system eliminates the friction of manual tape measurements. When coupled with generative image synthesis, it proposes a highly scalable, zero-hardware digital fitting room accessible via standard mobile devices.

18. Current Limitations

- Generative Artifacting: The Generative AI model occasionally produces visual artifacts or "hallucinations" when presented with highly complex human poses or heavily obscured source photos.

- Dependence on Accurate Input: The entire predictive pipeline assumes the user inputs truthful and accurate height and weight. Intentional or accidental misreporting results in cascading failures across measurements and sizing.

- Brand Extrapolation: The brand-sizing classifier is limited to the brands present in its training dataset. Unseen brands trigger a fallback to the less accurate standard sizing engine.